

UNIT 3: ADVANCED INTERNAL SEARCH METHODS

Residue-Based, Digital Data Structures, Spatial Organization, Indexing

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Course Outline

- 1 Residue-based Searches
- 2 Digital Data Structures
- 3 Spatial Organization Methods
- 4 In-Memory Indexing & Analysis

Linear
Binary
Hashing



Outline

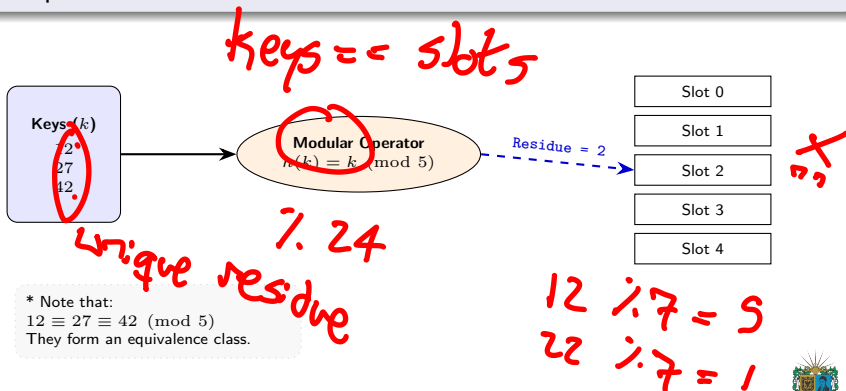
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Residue-Based Search Models I

Residue Search

An advanced retrieval framework using mathematical properties of modular arithmetic equivalences to map partitions of keys into distinct address spaces.



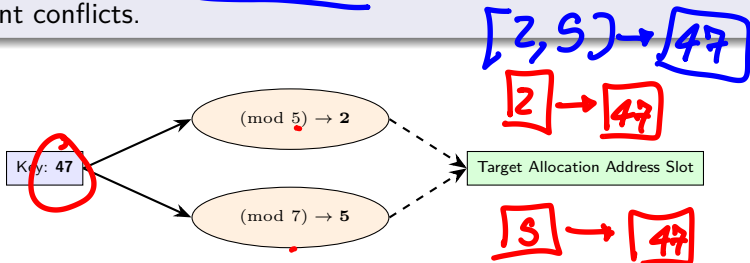
Residue-Based Search Models II

Multiple Residue Search

Improves simple key-to-slot mapping distributions by running the identifier through a cascade of distinct modular operations simultaneously.

$$H(k) = \{k \pmod{m_1}, k \pmod{m_2}\}$$

This multi-residue intersection pattern drastically reduces clustered placement conflicts.



Outline

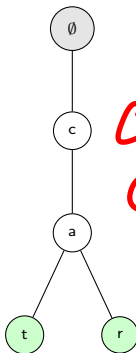
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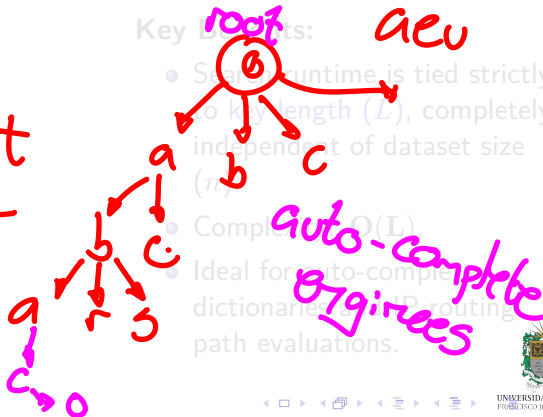
Digital Search Trees: Tries

Definition (Trie / Prefix Tree)

An **ordered tree data structure** where nodes *do not store* their actual search keys. Instead, a node's **position path in the tree hierarchy** dictates the associated **string index key value**.



Cat
Car



Key Facts:

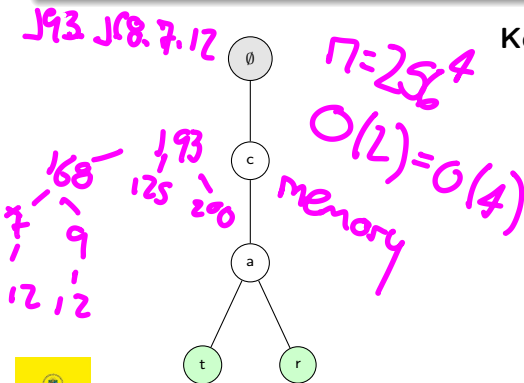
- Search runtime is tied strictly to key length (L), completely independent of dataset size (n)
- Complexity: $O(L)$
- Ideal for auto-complete dictionaries and IP routing path evaluations.



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Key Benefits:

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Spatial Search Methods

The Grid Method

Divides a *continuous multi-dimensional search coordinate space* into *uniform rectangular geometric cells (bins)*. Eliminates checking unrelated distant spaces.

2D Quad Trees

A space-partitioning data structure that alternates geometric axes coordinate boundaries (X-axis split, then Y-axis split) at each successive tree tier depth.



Cat, dog, mouse

meaning
+
context

Dict Cat 10
dog 15
mouse 45

embedding

10, x₁, y₁, z₁,
29, x₂, y₂, z₂,
48, x₃, y₃, z₃



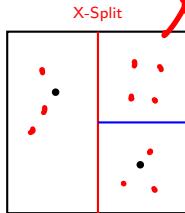
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A space-partitioning data structure that alternates geometric axis coordinate boundaries (**X-axis split**, then **Y-axis split**) at each successive tree tier depth.



operation
hard



Application Use Case: Range lookups, nearest neighbor geographic calculations, and multivariable correlation handling patterns



Spatial Search Methods

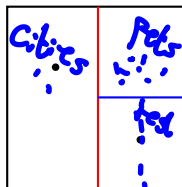
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cluster-function
X-Split



Y-Split



Geofence

Application Use Cases: Range lookups, nearest neighbor geographic calculations, and multi-variable collision handling patterns.



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In-Memory Index Tables

relational
databases

Index Table

A lightweight auxiliary structure containing ordered key fields alongside corresponding memory address references. Speeds up access without reorganizing massive raw underlying records.

binary &
search

10 26 30	3
11 11 20	2
12 14 16	0
14 14 20	1

0	1 2 4 16
1	14 14 20
2	11 11 20
3	10 26 30

students list
my data



Performance Metrics

Advanced Methods Efficiency Matrices

Data Structure	Average Search	Space Overhead	Dimensionality
Multiple Residue	$O(1)$ ✓	Medium (Sparse)	1D Unique Keys
Digital Tree (Trie)	$O(L)$ ✓	High (Pointers)	Sequential Strings
Grid Structure	$O(1)$ Avg ✓	High Array Scale	Multi-D Coordinates
k-d Spatial Tree	$O(\log n)$ ✓	Linear $O(n)$	k -Dimensions

Algorithmic Selection Rule

Choose **Tries** when working with string collections sharing prefix sub-patterns. Choose **k-d** trees when dealing with spatial bounds or dynamic geometric coordinates.

Special

Cases

to view



Performance Metrics

Advanced Methods Efficiency Matrices

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Nowadays: Index Table



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-ii>

